

# Pair of Linear Equations in Two Variables (Set A) — MCQ Worksheet

Mathematics • Chapter 3 • 25 Questions, 1 mark each — Total: 25 marks

**Instructions:** Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

**Q1.** A linear equation in two variables has the form:

- (A)  $ax + b = 0$
- (B)  $ax^2 + bx + c = 0$
- (C)  $ax + by + c = 0$
- (D)  $ax + by + cz = 0$

**Q2.** For lines  $a_1x + b_1y + c_1 = 0$  and  $a_2x + b_2y + c_2 = 0$  to be consistent and have unique solution:

- (A)  $a_1/a_2 = b_1/b_2 = c_1/c_2$
- (B)  $a_1/a_2 \neq b_1/b_2$
- (C)  $a_1/a_2 = b_1/b_2 \neq c_1/c_2$
- (D) Both equations are same

**Q3.** If  $a_1/a_2 = b_1/b_2 = c_1/c_2$ , the lines are:

- (A) Parallel
- (B) Coincident (infinitely many solutions)
- (C) Intersecting
- (D) Perpendicular

**Q4.** If  $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ , the lines are:

- (A) Coincident
- (B) Intersecting
- (C) Parallel (no solution)
- (D) Perpendicular

**Q5.** Geometrically,  $x - y = 2$  and  $x + y = 4$  are:

- (A) Parallel
- (B) Intersecting at (3, 1)
- (C) Coincident
- (D) Perpendicular only

**Q6.** Solution of  $x + y = 7$ ,  $x - y = 3$  is:

- (A) (2, 5)
- (B) (5, 2)
- (C) (3, 4)
- (D) (4, 3)

**Q7.** Solution of  $2x + y = 8$ ,  $x - y = 1$  is:

- (A) (3, 2)
- (B) (2, 3)
- (C) (1, 6)
- (D) (0, 8)

**Q8.** The pair  $x + 2y = 5$ ,  $3x + 6y = 15$  has:

- (A) No solution
- (B) Unique solution
- (C) Infinitely many solutions
- (D) Two solutions

**Q9.** The pair  $x + y = 3$ ,  $2x + 2y = 7$  has:

- (A) Unique solution
- (B) Infinitely many solutions
- (C) No solution
- (D) Exactly two

**Q10.** For  $3x - ky = 6$  and  $6x - 4y = 12$  to have infinitely many solutions,  $k =$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

**Q11.** The substitution method involves:

- (A) Substituting one variable in terms of another
- (B) Multiplying both equations
- (C) Comparing slopes
- (D) Cross-multiplication

**Q12.** Elimination method works by:

- (A) Removing one variable by adding/subtracting
- (B) Substituting
- (C) Drawing graph
- (D) Using cross multiplication formula

**Q13.** Cross-multiplication method gives  $x$  and  $y$  in terms of:

- (A)  $a_1, b_1, c_1$  etc
- (B) Sum of coefficients
- (C) Slopes
- (D) Roots

**Q14.** The sum of two numbers is 25 and difference is 7. The numbers are:

- (A) 16, 9
- (B) 18, 7
- (C) 20, 5
- (D) 15, 10

**Q15.** The cost of 5 pencils and 3 pens is Rs 35; cost of 3 pencils and 5 pens is Rs 37. Cost of 1 pencil and 1 pen is:

- (A) Rs 9
- (B) Rs 8
- (C) Rs 11
- (D) Rs 12

**Q16.** If 5 boys and 4 girls cost Rs 100 of fruit and 4 boys and 5 girls cost Rs 110, then per boy & per girl is:

- (A) Rs 8 and Rs 15
- (B) Rs 5 and Rs 10
- (C) Rs 10 and Rs 12.5
- (D) Rs 5 and Rs 18.75

**Q17.** For  $2x + 3y = 7$  and  $(a - b)x + (a + b)y = 3a + b - 2$  to have infinitely many solutions:

- (A)  $a = 5, b = 1$
- (B)  $a = 1, b = 5$
- (C)  $a = 2, b = 3$
- (D)  $a = 0, b = 0$

**Q18.** Two lines are coincident if they have:

- (A) Different slopes
- (B) Same slope but different intercept
- (C) Same slope and same intercept
- (D) Slopes adding to 0

**Q19.** If we plot  $2x + 3y = 12$  and  $4x + 6y = 24$ , lines are:

- (A) Parallel
- (B) Coincident
- (C) Intersecting
- (D) Perpendicular

**Q20.** Half the perimeter of a rectangular garden is 36 m. If length is 4 m more than width, the dimensions are:

- (A)  $20 \text{ m} \times 16 \text{ m}$
- (B)  $18 \text{ m} \times 14 \text{ m}$
- (C)  $22 \text{ m} \times 14 \text{ m}$
- (D)  $16 \text{ m} \times 12 \text{ m}$

**Q21.** In a fraction, if numerator is reduced by 1 it becomes  $\frac{1}{3}$ , and if denominator is increased by 8 it also becomes  $\frac{1}{3}$ . The fraction is:

- (A)  $\frac{3}{8}$
- (B)  $\frac{2}{5}$
- (C)  $\frac{5}{12}$
- (D)  $\frac{4}{12}$

**Q22.** A boat goes 30 km upstream in 5 h and 30 km downstream in 3 h. Speed of stream is:

- (A) 1 km/h
- (B) 2 km/h
- (C) 3 km/h
- (D) 4 km/h

**Q23.** The age of father is 3 times that of son. After 12 years it will be twice. Son's present age is:

- (A) 10
- (B) 12
- (C) 14
- (D) 16

**Q24.** For unique solution we need:

- (A) Determinant of coefficient matrix non-zero ( $a_1b_2 - a_2b_1 \neq 0$ )
- (B)  $a_1 = a_2$
- (C)  $b_1 = b_2$
- (D)  $c_1 = c_2$

**Q25.** Lines having same y-intercept can still:

- (A) Have different slopes
- (B) Always be coincident
- (C) Always be parallel
- (D) Always meet at origin